

One-Minute Video Generation with Test-Time Training

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Learning to (Learn at Test Time): RNNs with Expressive Hidden States

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Introduction

- Limitations

1. Can make only short clips

- The maximum length of public APIs for video generation

- Sora (OpenAI): 20 secs
- MovieGen (Meta): 16 secs
- Ray 2 (Luma): 10 secs
- Veo2 (Google): 8 secs

2. Can make only single scenes without complex stories(~March 2025)

Introduction

- Limitations
 - one-minute videos requires over 300k tokens in context
 - Transformers increases quadratically (actually linear with KV cache) with context length
 - generating a one-minute video
 - 11× longer than generating 20 videos of 3 seconds each
 - training would have taken 12× longer

Introduction

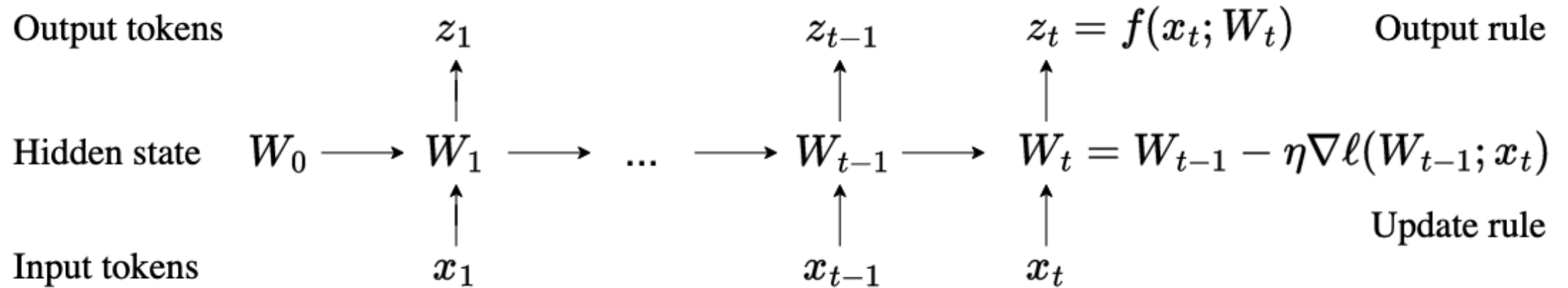
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Introduction

- Solution
 - To address this prob, using RNN (especially linear attention) as alternative
 - E.g. mamba, DeltaNet
 - But contain only single scenes, slow motion and simple stories
 - Because RNN architecture has less expressive (fixed) hidden states
 - E.g. cell state, A matrix
 - Replace linear (matrix) to **two-layer MLP with 2 × more hidden cells** (=> 4 x hidden, richer nonlinearities)

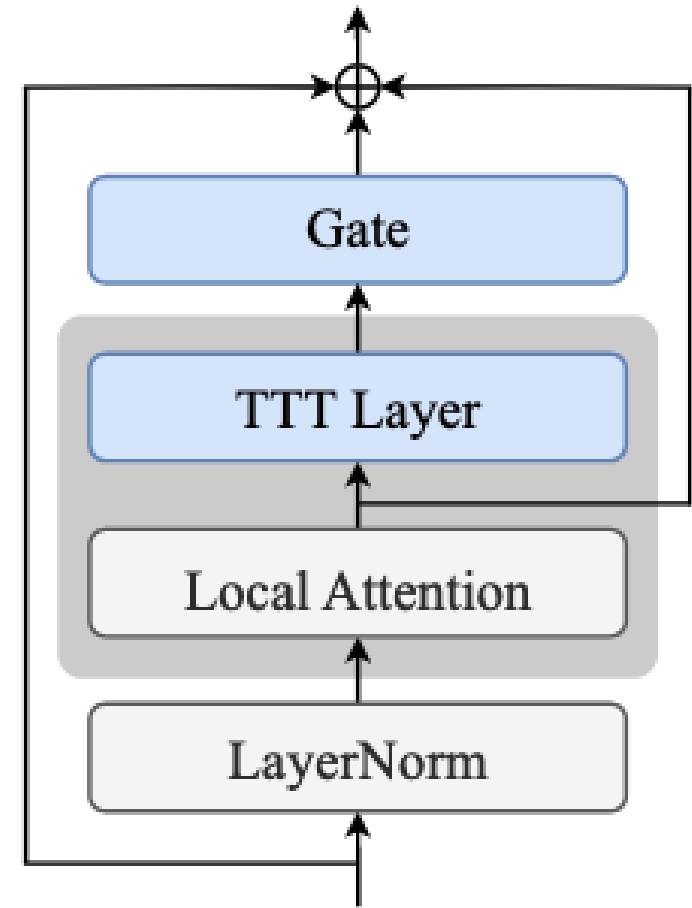
Introduction

- $W \rightarrow \text{MLP} * 2$



Introduction

- Base model: CogVideo-X 5B (DiT)
- Add TTT-layer -> use as long term memory like RNN
- + Training env: 50 hour 256 H100..



TTT as Updating a Hidden State

- We will use self-supervised learning to compress context $x_1, \dots, x_t$ into a hidden state W_t
- Why SSL?
 - Storing contexts as a large buffer is neither space nor computationally feasible
 - So, see context as unlabeled dataset
 - Can't use supervised -> use SSL
- Also, can train even at test time by using unlabeled dataset

TTT as Updating a Hidden State

- TTT-layer can be trained only at test time

$$W_t = W_{t-1} - \eta \nabla \ell(W_{t-1}; x_t)$$

$$\ell(W; x_t) = \|f(\tilde{x}_t; W) - x_t\|^2$$

$$z_t = f(x_t; W_t)$$

Learning a Self-Supervised Task for TTT

- Instead of handcrafting a SS task from human priors
- more end-to-end approach, learning it as part of train

$$\tilde{x}_t = \theta_K x_t$$

$$\ell(W; x_t) = \|f(\tilde{x}_t; W) - x_t\|^2$$



$$\ell(W; x_t) = \|f(\theta_K x_t; W) - \theta_V x_t\|^2$$

$$z_t = f(x_t; W_t)$$



$$z_t = f(\theta_Q x_t; W_t)$$

TTT-MLP Instantiation

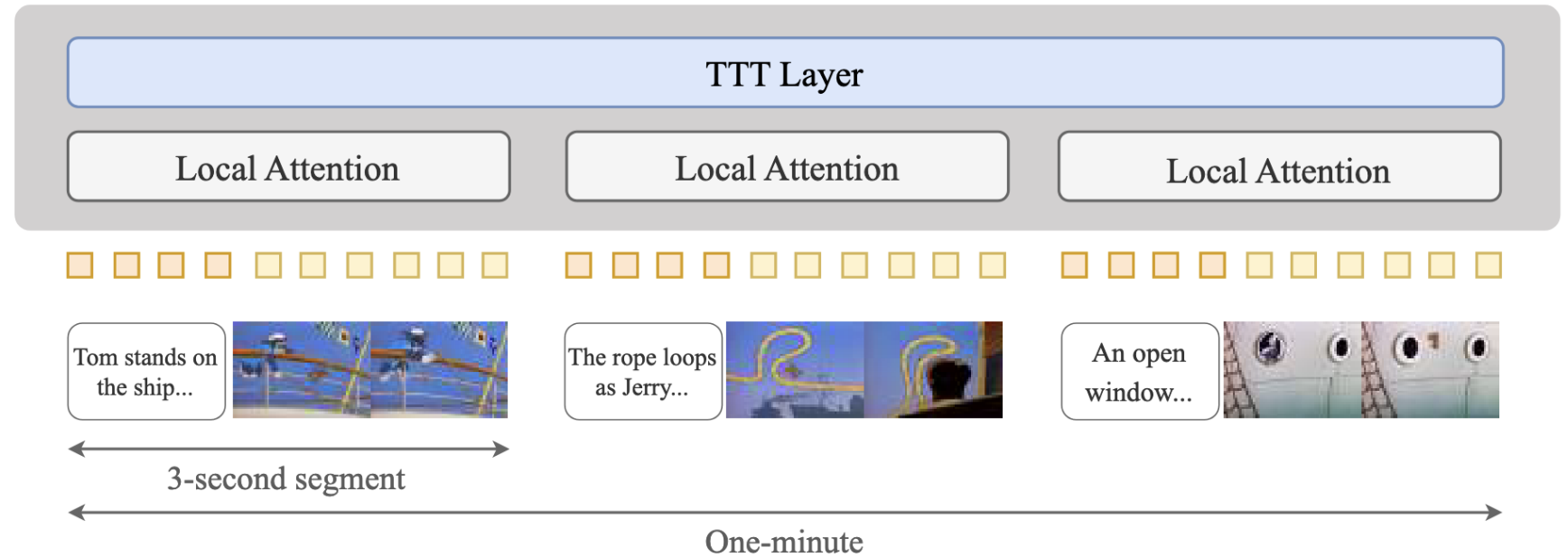
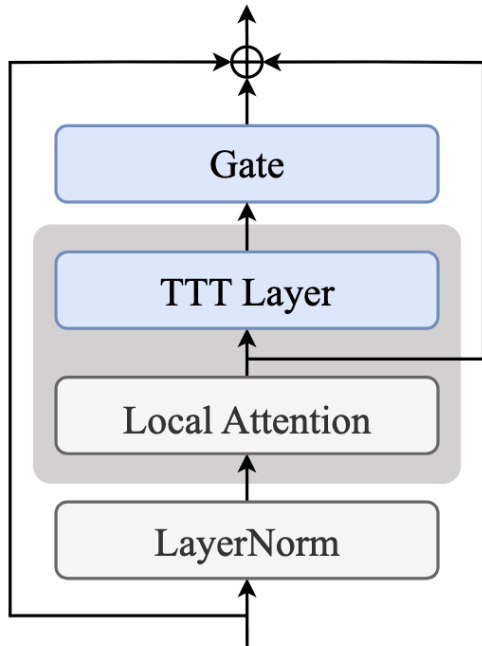
- $f(x)$ is called TTT-MLP

$$f(x) = x + \text{LN}(f_{\text{MLP}}(x))$$

- $f_{\text{MLP}}(x)$: a two-layer MLP (hidden dimension is $4 \times$ the input dimension)
- Use GELU as activation function

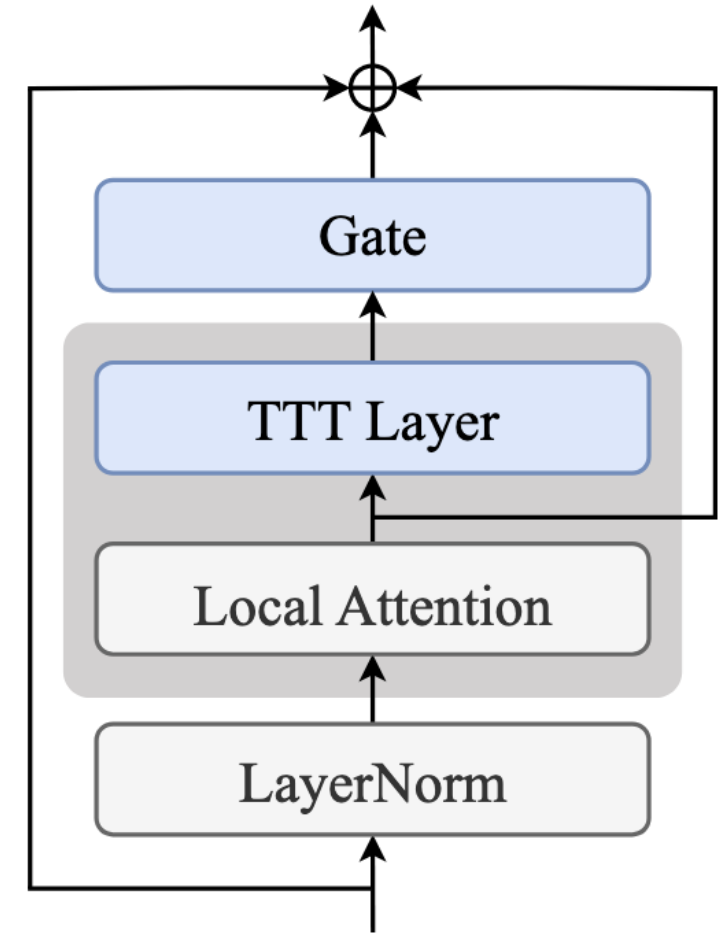
Architecture

- Overview



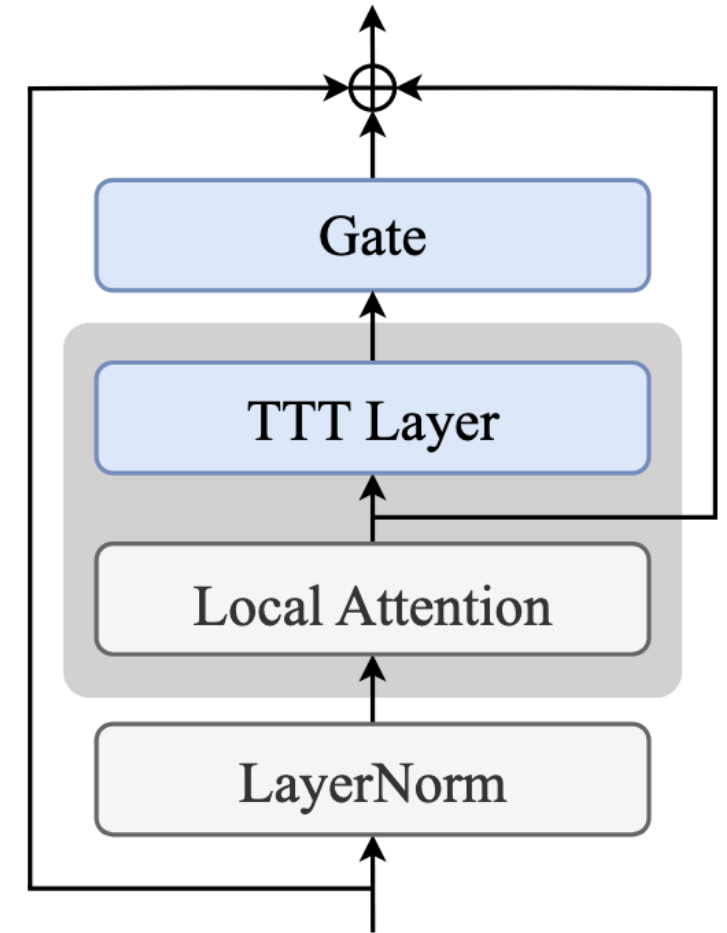
TTT layer with a learnable gate

- Gating
- Input sequence: $X = (x_1, \dots, x_t)$, $x_t \in \mathbb{R}^d$
- Output Sequence: $Z = (z_1, \dots, z_t) = TTT(X)$, $z_t \in \mathbb{R}^d$
- $gate(TTT, X; \alpha) = \tanh(\alpha) \otimes TTT(X) + X$
- $\tanh(\alpha) \in (-1, 1)^d$



TTT layer with a learnable gate

- $X' = self_attn(LN(X))$
- $Z = gate(TTT, X'; \alpha)$
- $Z' = gate(TTT', Z; \beta)$
- $\left(TTT' = rev\left(TTT(rev(X)) \right), rev(X) = (x_T, \dots, x_1) \right)$
- $Y = Z' + X$



Sequence Example

$$V_{\text{noise},t}^1 = [x_t^{1,1}, x_t^{1,2}, \dots, x_t^{1,1152}] \in \mathbb{R}^{d \times 1152}$$

$$V_{\text{noise},t}^2 = [x_t^{2,1}, x_t^{2,2}, \dots, x_t^{2,1152}] \in \mathbb{R}^{d \times 1152}$$

$$E_{\text{text}}^1 \in \mathbb{R}^{d \times 30} \quad (e_1^1, \dots, e_{30}^1)$$

$$V_{\text{noise},t}^1 \in \mathbb{R}^{d \times 1152} \quad (x_t^{1,1}, \dots, x_t^{1,1152})$$

$$E_{\text{text}}^2 \in \mathbb{R}^{d \times 40} \quad (e_1^2, \dots, e_{40}^2)$$

$$V_{\text{noise},t}^2 \in \mathbb{R}^{d \times 1152} \quad (x_t^{2,1}, \dots, x_t^{2,1152})$$

$$\text{SegmentSeq}_t^2 = [e_1^2, e_2^2, \dots, e_{40}^2, x_t^{2,1}, x_t^{2,2}, \dots, x_t^{2,1152}]$$

$$\text{FullInputSeq}_t = \left[\underbrace{\text{SegmentSeq}_t^1}_{1182\text{토큰}}, \underbrace{\text{SegmentSeq}_t^2}_{1192\text{토큰}} \right]$$



Fine-Tuning Recipe and Dataset

- Multi-stage context extension
 - First stage: full-tuning(adaptation), 3 secs
 - The others: TTT layers and gates, 9 -> 18 -> 30 -> 63

Experiments



TTT-MLP preserves temporal consistency over scene changes and across angles, producing smooth, high-quality actions.



Sliding-window attention alters the kitchen environment, changes the house color, and duplicates Jerry stealing the pie.



Gated DeltaNet lacks temporal consistency across different angles of Tom but maintains the kitchen environment in later frames.



Mamba 2 distorts Tom's appearance as he growls and chases Jerry but maintains a similar kitchen environment throughout the video.

Experiments

	Text following	Motion naturalness	Aesthetics	Temporal consistency	Average
Mamba 2	985	976	963	988	978
Gated DeltaNet	983	984	993	1004	991
Sliding window	1016	1000	1006	975	999
TTT-MLP	1014	1039	1037	1042	1033

Table 1. Human evaluation results for one-minute videos. TTT-MLP improves over the second best method by 34 Elo points on average. Axes with the most improvements are scene consistency (+38) and motion smoothness (+39). For context, GPT-4 scores 46 Elo points over GPT-3.5 Turbo, and GPT-4o scores 29 over GPT-4 Turbo in Chatbot Arena [6].

Experiments



Temporal consistency: The boxes morph between 3-second segments of the same scene.



Motion naturalness: The cheese hovers in mid-air rather than falling naturally to the ground.



Aesthetics: The lighting in the kitchen becomes dramatically brighter as Tom turns around.

Limitations

- Short context
 - TTT-Linear > TTT-MLP for 18 secs
- Wall-clock time
 - inference and training with TTT-MLP are $1.4\times$ and $2.1\times$ slower than with Gated DeltaNet
- Video artifacts
 - still contain notable artifacts, especially in motion naturalness and aesthetics.
 - CogVideo-X also seem to have limited motion naturalness and aesthetics

Thank You!